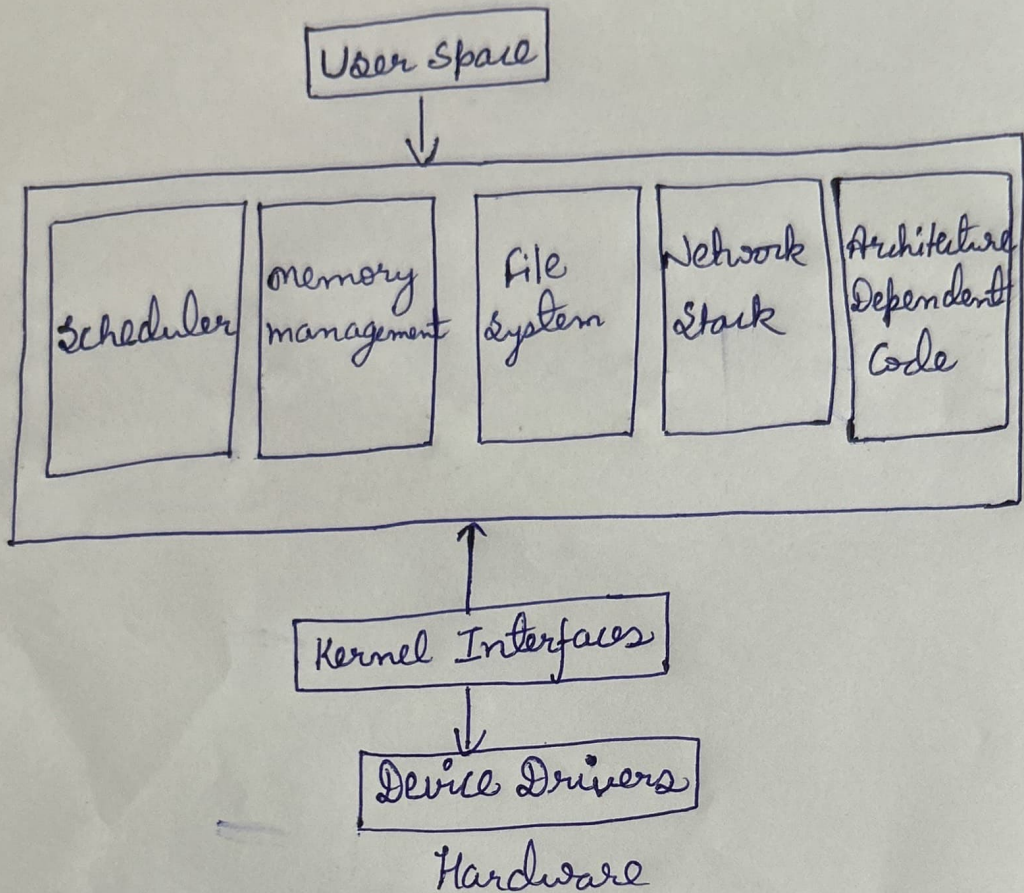


LSP - Task-1

Linux Kernel Architecture



- i) Scheduler :- The scheduler is incharge of deciding which process runs on the CPU and for how long. It handles process prioritization and context switching to ensure efficient multitasking.
- ii) Memory Management :- This component manages how memory is allocated and freed for application and processes. It includes mechanisms like virtual memory and paging to optimize RAM usage.
- iii) File System :- The file system organizes and controls access to data stored on devices. It handles file reading, writing, permission and supports different file formats.

(iv) Network Stack:- This part deals with all networking operations. It enables communication between devices using various protocols like TCP/IP, ensuring data is sent and received properly.

(v) Architecture - Dependent Code:- These are the hardware-specific parts of the kernel. They let the same kernel work on different processors and platforms by adapting to their unique features.

(vi) Kernel Interfaces:- These are the links between user applications and the kernel, including system calls and libraries. They let programs request kernel-level services.

(vii) Device Drivers:- Drivers act as translators between hardware and the rest of the system. They allow the kernel to communicate with devices like keyboard, disks and network cards.